

P2P – PEER-TO-PEER FILE SHARING APP

NODE.JS | HTML / CSS | COMPUTER NETWORKING | OPERATING SYSTEMS

TEAM MEMBERS

- A.KARTHIK
- B.MUKESH
- Y.SHIVAPRASAD
- K.YADAGIRI
- S.JEEVAN

PROBLEM STATEMENT:

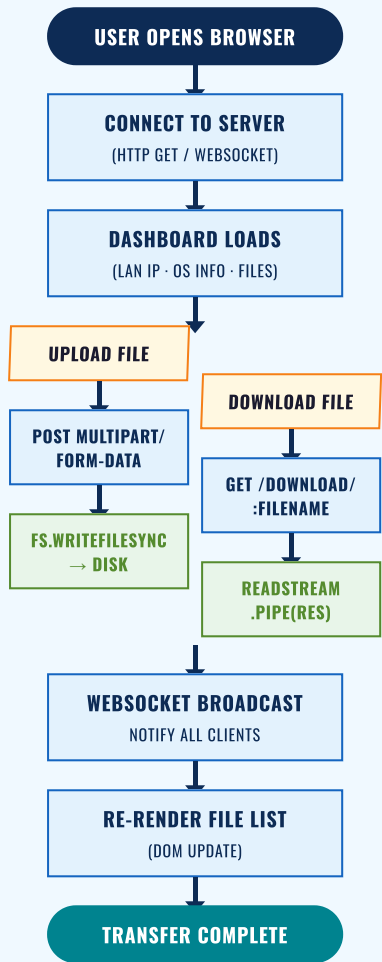
Existing file-sharing methods depend on cloud platforms, internet access, or proprietary software installation. Within a Local Area Network (LAN), there is no universal, browser-accessible tool that works on any device instantly. The absence of a real-time, self-hosted solution makes quick file exchange within institutions, labs, and offices inefficient and insecure.

GUIDED BY FACULTY

- D.N.B.T SUNDARI– CN
- A.SANDHYA RANI– NODEJS
- B.SURYA NARAYANA– OS

SYSTEM DESIGN / FLOW DIAGRAM:

Flow Chart: P2P File Share System



ABSTRACT:

P2P File Sharing App is a browser-based solution that enables users on the same **Local Area Network (LAN)** to upload, share, and download files without cloud or internet dependency. Built using **Node.js** (HTTP + WebSocket server), **HTML5 & CSS3** (drag-and-drop UI), the system demonstrates **Computer Networking** concepts (TCP/IP, OSI model, WebSocket, multipart/form-data) and **OS concepts** (file system I/O, system calls, memory-efficient streaming via `fs.createReadStream`).

FEATURES

- LAN-Based File Sharing
- Drag-and-Drop Upload (up to 500 MB)
- Real-Time Updates via WebSocket
- Path Traversal Security Guard
- Streaming Downloads (Memory Efficient)
- OS Info Dashboard (CPU, RAM, IP)
- Live Activity Log Panel

ADVANTAGES:

- ✓ Zero cloud dependency — works offline on LAN
- ✓ No installation required on client devices
- ✓ Memory-efficient streaming (constant ~64 KB RAM)
- ✓ Single npm dependency — only the ws package
- ✓ Simultaneous multi-client WebSocket support
- ✓ Cross-platform — Windows, Linux, macOS
- ✓ Sub-second file transfer initiation (< 300 ms)

CONCLUSION:

The P2P File Sharing App successfully addresses LAN-based file transfer by providing:

- Complete Protocol Coverage** — HTTP, WebSocket, TCP/IP, and multipart encoding all in one project
- Real-Time Communication** through full-duplex WebSocket broadcasting to all LAN clients
- OS-Level Integration** with live CPU, RAM, and network interface data via Node.js os module
- Secure Platform** with path sanitization preventing path traversal attacks
- Accessible Solution** — responsive web UI works on phones, tablets, and desktops over Wi-Fi

Future Enhancements

- HTTPS / WSS Encryption** — TLS certificates for encrypted file transfers over the network
- QR Code Connect** — Auto-generate QR code of LAN URL for instant mobile access
- Node.js Cluster Mode** — Multi-core CPU utilization using cluster module for high load
- User Authentication** — Token-based login so only authorized users can upload or delete
- Cloud Deployment** — Docker containerization for AWS / Google Cloud hosting